# Project Proposal

Project name: Pet Macher

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# What?

Every year, millions of dogs and cats end up in shelters. Sadly, not all of them ever find a home. For example, in 2023 alone, more than 6.5 million animals entered shelters in the U.S., and hundreds of thousands never made it out. In Europe, shelters face the same challenges - thousands of animals waiting for someone to give them a second chance. The truth is, many pets are returned simply because they weren't the right fit for their new families.

This project, Pet Matcher, is an AI-powered tool designed to change that. The idea is simple: just like dating apps match people, Pet Matcher matches people with pets. Future owners can choose the things that matter to them - whether it's the type of animal, breed, size, color, age, or even personality traits. The system then looks at shelter data, such as temperament, activity level, and special needs, to recommend the pets that fit best.

The goal is simple: help people find pets that truly fit into their lives, and help pets finally find a loving forever home.

# Why?

Because adoption should be about creating a bond, not just picking the first cute face you see.

- Shelters are overcrowded, and too many pets are overlooked.
- Families often choose pets that don't fit their lifestyle, which leads to disappointment and returns.
- Every failed adoption is tough - not only for the family, but especially for the pet who goes back to the shelter.

By making adoption smarter and more personalized, Pet Matcher can increase successful adoptions, reduce shelter overcrowding, and give more animals a real chance at happiness.

# Who?

- **Primary users:** People looking to adopt a pet who want guidance in finding the right fit.
- **Secondary users:** Shelters and rescue organizations, who benefit from faster and more successful adoptions.
- **Other stakeholders:** Veterinarians, volunteers, and trainers who provide data about pets' health and behavior.

**In short:** it's for anyone who cares about making adoption easier and more successful - for both people and animals.

# When?

## Phase 1: Proposal (Week 1)

- **Domain Understanding:** Research the adoption problem, look into shelter challenges, and understand user needs (adopters and shelters).
- **Analytic Approach:** Define how AI will be used (recommendation system, NLP for preferences, scoring model).
- **Data Requirements:** Identify what data is needed (pet traits, adopter lifestyle questionnaires, adoption outcome records).

## Phase 2: Provisioning (Weeks 2–3)

**Week 2 – Data Collection & Understanding**

- Gather open datasets (PetFinder).
- Collect sample shelter profiles (species, breed, age, personality).
- Create adopter lifestyle questionnaire (activity level, home type, allergies, etc.).
- Explore the data - find missing values, common patterns.

**Week 3 – Data Preparation**

- Clean and preprocess pet data (normalize traits, encode categories like temperament).
- Prepare adopter input data structure.
- Split data into training/testing sets if needed.

## Phase 3: Prediction (Weeks 4–5)

**Week 4 – Preprocessing & Modeling**

- Build first matching prototype (similarity scoring or recommendation engine).
- Add adopter input questionnaire → generate first match results.
- Try simple models (rule-based or ML classification).

**Week 5 – Evaluation**

- Test with users or simulated data.
- Measure how accurate/reasonable the matches are.
- Adjust algorithm for better compatibility scoring.

## Phase 4: Delivery (Week 6)

- **Demonstration:** Build a simple web/app interface where users can enter preferences and see matches.
- **Feedback:** Share with classmates, mentors, or test users for feedback.
- **Next Steps:** Plan improvements (partner with shelters, add NLP for free-text input, expand dataset).

# How?

**Pet Matcher would work in a few simple steps:**

- The adopter fills out a short quiz about their lifestyle (active or calm, kids at home, size of living space).
- The system uses AI to compare those answers with shelter data (breed, temperament, energy level, age, etc.).
- It shows a list of pets with compatibility scores - like “This dog matches you 85%.”
- Adopters can learn why a certain pet is a good fit and contact the shelter directly.

**Behind the scenes, the AI combines:**

- Recommendation systems to find the best matches.
- Natural language processing so people can describe what they want in simple words ( “I want a playful dog that’s good with kids”).
- Optional image recognition to help shelters identify breeds or traits from photos.